

Algebra II with Trigonometry

Chapter 5.3

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:

[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Problems

1. Two triangles have the same area. The first triangle has side lengths of 5 and 8 with an included angle of 15° . The second triangle has side lengths of 4 and 10 with an included obtuse angle θ . What is the degree measure of θ ?
2. Given that $\sin \theta + \cos \theta = \frac{1}{5}$ and that θ lies in Quadrant II, determine the exact value of $\sin \theta - \cos \theta$.
3. Find the exact sum of the first 100 terms of the following sequence:
 $\cos(120^\circ) + \cos(240^\circ) + \cos(360^\circ) + \cos(480^\circ) + \dots$
4. The six vertices of a regular hexagon centered at the origin correspond to standard position angles $\theta_1, \theta_2, \dots, \theta_6$. If one of the vertices lies on the positive x -axis, mentally evaluate the product:
 $(1 - \sin \theta_1)(1 - \sin \theta_2)(1 - \sin \theta_3)(1 - \sin \theta_4)(1 - \sin \theta_5)(1 - \sin \theta_6)$
5. A circular sector has an acute central angle θ . The ratio of the area of the triangle formed by its two bounding radii and the chord connecting their endpoints, to the area of the entire sector, is exactly $\frac{3}{\pi}$. Find the degree measure of θ .
6. If θ is an acute angle in standard position such that $\sec \theta + \tan \theta = 4$, compute the exact value of $\sin \theta$.